Abstract

No prior automated system tracks hypothesis-level evidence across the full Active Inference and Free Energy Principle (FEP) literature. Manual synthesis cannot keep pace with a field that has grown at a compound annual rate of 16.99% across 2005–2026, and the FEP's theoretical generality has invited falsifiability critiques colombo2021free that only hypothesis-specific evidence profiling can address. Building on the systematic literature analysis of Knight, Cordes, and Friedman knight2022fep—which pioneered manual annotation paired with ontology-based analysis at the scale of hundreds of papers—we present a computational meta-analysis framework that automates and scales this approach.

The pipeline retrieves literature from arXiv, Semantic Scholar, and OpenAlex, deduplicating $N = 849$ papers via a canonical identifier hierarchy (DOI $>$ arXiv ID $>$ Semantic Scholar ID $>$ OpenAlex ID). It classifies papers into a three-tier taxonomy spanning eight categories: A (Core Theory), B (Tools & Translation), and C (Application Domains). An LLM-powered extraction system then evaluates each abstract against eight core hypotheses, producing